SET-A

NIZWA COLLEGE OF TECHNOLOGY
DEPARTMENT OF INFORMATION TECHNOLOGY

Final Examination

Semester 3, Academic Year 2017-2018

Course Code: MATH1200 Course Title: CALCULUS - 1

Date: 01-07-2018 **Time:** 12 NN – 2 PM

Max Marks	50	Time Allocation	2 Hours
Student ID		Student Name	
Section		Level	DIPLOMA

Instructions for students

1. Exam paper consists of 8 pages including cover page.
2. Answer the questions in the space provided by using **blue or black pen** only.
3. Show the steps for the answers clearly.
4. Calculator is allowed.
5. Write the correct units, if any.

Question Number → PART ↓	1	2	3	4	5	6	7	8	9	10	Total
PART A (MCQ) (10 × 1 = 10 Marks)											
PART B (7 × 4 = 28 Marks)											
PART C (2 × 6 = 12 Marks)											
Total											

Question Setter	Moderator	HOS
	24/6/18.	24/6/2018

Course Tutors

1) 21/6/18. 3) 24/6 4. Manikta Riji

2) 21/6/18

PART A – MULTIPLE CHOICE QUESTIONS

Answer all questions. Each question carries 1 mark.

A1. Given $f(x)=\frac{4}{x^2}$. The value of $f'(2)$ is _____.

(a) -2 (b) -1 (c) 1 (d) 2

A2. $\int_0^4 \frac{1}{2\sqrt{x}}\,dx =$ _____.

(a) 1 (b) $\sqrt{8}$ (c) 2 (d) 4

A3. At time t sec, the position of a body moving on a coordinate plane is $s=t^3-6t^2+9t$ meters. The body moves backward in the interval _____.

(a) $(1,3)$ (b) $(0,1)$ (c) $(3,\infty)$ (d) $(0,\infty)$

A4. The curve $y=x+\frac{9}{x}$ has a horizontal tangent at the point _____.

(a) $(0,0)$ (b) $(3,-6)$ (c) $(-3,9)$ (d) $(3,6)$

A5. The derivative of the function $f(x)=e^{3\ln x}$ with respect to x is _____.

(a) $e^{3\ln x}$ (b) $e^{\frac{3}{x}}$ (c) $3x^2$ (d) $3x$

A6. The absolute maximum value of $f(x)=4-x^2$ in the interval $-3\le x\le 1$ is _____.

(a) -5 (b) 0 (c) 1 (d) 4

A7. If $r=\sin\left(\frac{3\pi t}{2}\right)$, then $\frac{dr}{dt}=$ _____.

(a) $\sin\left(\frac{3\pi}{2}\right)$ (b) $\frac{3\pi}{2}\cos\left(\frac{3\pi t}{2}\right)$ (c) $\frac{3}{2}\cos\left(\frac{3\pi t}{2}\right)$ (d) $-\frac{3}{2}\cos\left(\frac{3\pi t}{2}\right)$

A8. If $x=\sin y$, then $\frac{dy}{dx}=$ _____.

(a) $\sec y$ (b) $-\csc y$ (c) $\cos y$ (d) $-\sin y$

A9. $\int \frac{1}{\sec x}\,dx=$ _____.

(a) $\sec x$ (b) $\cos x$ (c) $\sec x\tan x$ (d) $\sin x$

A10. $\frac{d}{dx}\left(\sin^{-1}2x\right)=$ _____.

(a) $\frac{-1}{\sqrt{1-2x^2}}$ (b) $\frac{1}{\sqrt{1-4x^2}}$ (c) $\frac{2}{\sqrt{1-4x^2}}$ (d) $\frac{-2}{\sqrt{1-4x^2}}$

PART B – SHORT ANSWER QUESTIONS

Answer all questions. Each question carries 4 marks.

B1. If $x = 2\cos t$, $y = 2\sin t$, find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ at $t = \frac{\pi}{4}$.

$$\frac{dx}{dt} = -2\sin t \quad (0.5) \qquad \frac{dy}{dt} = 2\cos t \quad (0.5)$$

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{2\cos t}{-2\sin t} = -\cot t. \quad (1)$$

$$\left.\frac{dy}{dx}\right]_{t=\pi/4} = \frac{dy}{d} - \cot \pi/4 = \underline{\underline{-1}} \quad (0.5)$$

$$\frac{d^2y}{dx^2} = \frac{\left(\frac{dy}{dx}\right)'}{\frac{dx}{dt}} = \frac{(-\cot t)'}{-2\sin t} = \frac{\csc^2 t}{-2\sin t} = -\frac{1}{2}\csc^3 t \quad (1)$$

$$\left.\frac{d^2y}{dx^2}\right]_{t=\pi/4} = -\frac{1}{2}\csc^3 \pi/4 = \underline{\underline{-\sqrt{2}}} \quad (0.5)$$

B2. Find the derivative of $y = \tan\left(\frac{\sin x}{x}\right)$ with respect to x, by using the Chain Rule.

$$\frac{dy}{dx} = \sec^2\left(\frac{\sin x}{x}\right)\left[\frac{(\sin x)'x - (\sin x)x'}{x^2}\right] \rightarrow (3)$$

$$= \sec^2\left(\frac{\sin x}{x}\right)\cdot\left[\frac{x\cos x - \sin x}{x^2}\right] \quad \cdot (1)$$

B3. The acceleration of a particle moving on a coordinate line is $a = -4\sin 2t$. If its initial velocity $v(0) = 2$ and initial position $s(0) = -3$, then find the body's position at time t.

$a = \frac{dv}{dt} = -4\sin 2t$ (0.5)

$\int dv = \int -4\sin 2t \, dt$

$v = 2\cos 2t + c$ (1)

$v(0) = 2 \Rightarrow 2 = 2(1) + c$

$\Rightarrow c = 0$ (0.5)

$\therefore v = 2\cos 2t$

$v = \frac{ds}{dt} = 2\cos 2t$ (0.5)

$\int ds = \int 2\cos 2t \, dt$

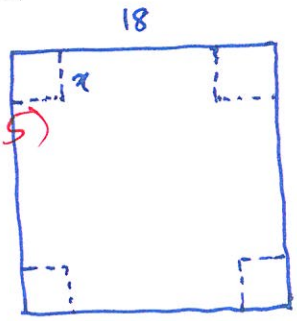
$s = \sin 2t + c'$ (1)

$s(0) = -3 \Rightarrow -3 = 0 + c'$

$\Rightarrow c' = -3$

$\therefore s = \sin 2t - 3$. (0.5)

B4. An open box is to be made from a 18 inch by 18 inch sheet of cardboard by cutting congruent squares out of the four corners and folding up the sides. What dimensions of the box will yield the largest volume? What is the maximum volume of the box?



Dimensions of the box, Length $L = 18 - 2x$
Width $W = 18 - 2x$ → (0.5)
Height $H = x$

Volume of the box, $V = LWH$

$= (18 - 2x)(18 - 2x)x$

$= 4x^3 - 72x^2 + 324x$ → (1)

$V' = 0 \Rightarrow 12x^2 - 144x + 324 = 0$ → (1)

$\Rightarrow x = 3, 9.$

At $x = 9$, $V = 0$ as $L = 0, W = 0, H = 9$.

At $x = 3$, $L = 12$, $W = 12$, $H = 3$, $V = 12 \times 12 \times 3 = 432$.

$\therefore$ Volume is maximum when $L = W = 12$ inch, $H = 3$ inch → (1)

maximum volume $= 432$ inch^3. → (0.5)

B5. The voltage V (volts), current I (amperes) and resistance R (ohms) of an electric circuit are related by the equation $V = IR$. Suppose that V is increasing at the rate of 1 volt/sec while I is decreasing at the rate of $\frac{1}{3}$ amp/sec. Let t denote the time in seconds.

(a) What are the values of $\frac{dV}{dt}$ and $\frac{dI}{dt}$?

(b) What equation relates $\frac{dR}{dt}$ to $\frac{dV}{dt}$ and $\frac{dI}{dt}$?

(c) Find the rate at which R is changing when $V = 12$ volts and $I = 2$ amp.

(a) $\frac{dV}{dt} = 1$ volt/sec , $\frac{dI}{dt} = -\frac{1}{3}$ amp/sec (0.5 + 0.5)

(b) $V = IR \Rightarrow R = \frac{V}{I}$.

$$\therefore \frac{dR}{dt} = \frac{\frac{dV}{dt} \cdot I - V \cdot \frac{dI}{dt}}{I^2}$$ (2)

(c) $$\left.\frac{dR}{dt}\right]_{V=12,\, I=2} = \frac{(1)(2) - (12)(-\frac{1}{3})}{(2)^2}$$

$$= \frac{2+4}{4} = \frac{6}{4} = \frac{3}{2} \text{ ohms/s}$$ (1)

B6. Evaluate the integral $\int \left(\frac{1}{x^2} - 3x^2 + \frac{5}{x} - 9 \right) dx$.

$$\int \left(\frac{1}{x^2} - 3x^2 + \frac{5}{x} - 9 \right) dx = -\frac{1}{x} - x^3 + 5\ln x - 9x + C$$

(1) (1) (1) (1)

B7. Given the curve $x^2 + y^2 = 25$ and the point $(-4,3)$.

(a) Verify that the given point is on the curve.

(b) Find the equation of the tangent to the curve at the given point.

(c) Find the equation of the normal to the curve at the given point.

(a) $x^2 + y^2 = 25, \quad (-4, 3)$

$(-4)^2 + (3)^2 \quad | \quad 25$

$16 + 9$

$25 \longleftrightarrow 25$ LHS = RHS

$\therefore$ Point lies on the curve. (1)

(b) $x^2 + y^2 = 25$

taking derivative w.r.t x,

$2x + 2y\frac{dy}{dx} = 0$

$\frac{dy}{dx} = -\frac{x}{y}$ (1)

Slope, $m = \left.\frac{dy}{dx}\right]_{(-4,3)} = \frac{-(-4)}{3} = \frac{4}{3}$. (0.5)

Equation of tangent at $(-4,3)$ is

$y - 3 = \frac{4}{3}(x + 4)$ (0.5)

(c) Slope of normal at $(-4,3) = -\frac{1}{m} = -\frac{3}{4}$. (0.5)

Equation of normal at $(-4,3)$ is

$y - 3 = -\frac{3}{4}(x + 4)$ (0.5)

PART C – LONG ANSWER QUESTIONS

Answer all questions. Each question carries 6 marks.

C1. For the function $f(x) = 2x^3 - 6x^2 + 12$

(a) Find the intervals in which the function is increasing or decreasing.

(b) Find the intervals in which the function is concave up or concave down.

(c) Find the local extreme values and inflection points of the function.

(a) $f'(x) = 6x^2 - 12x$

$f'(x) = 0 \Rightarrow 6x^2 - 12x = 0$

$6x(x-2) = 0 \Rightarrow x = 0, 2.$

	$(-\infty, 0)$	$(0, 2)$	$(2, \infty)$
sign of $f'(x)$	+	−	+
behaviour of f	f is increasing	f is decreasing	f is increasing

~~f has~~

→ (2)

(b) $f''(x) = 12x - 12.$

$f''(x) = 0 \Rightarrow 12x - 12 = 0 \Rightarrow x = 1$

	$(-\infty, 1)$	$(1, \infty)$
sign of $f''(x)$	−	+
behaviour of f	f is concave down	f is concave up.

→ (2)

(c) From table 1, f has local maximum at $x = 0$ and local minimum at $x = 2$.

Local maximum value $= f(0) = 12$

Local minimum value $= f(2) = 4$

→ (1)

Inflection point occurs at $x = 1$. $f(1) = 8$

$\therefore$ Inflection point $= (1, 8)$ → (1)

C2. A dynamite blast blows a heavy rock straight up in the air with a launch velocity of 192 feet/sec. It reaches a height of $s = 192t - 16t^2$ feet after t seconds.

(a) How high does the rock go?

(b) What are the velocity and speed of the rock when it is 512 feet above the ground on the way up? On the way down?

(c) How long is the rock aloft?

(a) At maximum height, velocity = 0.

$$v = \frac{ds}{dt} = 192 - 32t.$$

$$v = 0 \Rightarrow 192 - 32t = 0$$

$$t = 6 \text{ sec} \rightarrow (1)$$

$\therefore$ Maximum height, $s_{max} = 192(6) - 16(6)^2 = 576$ feet. (1)

(b) Height = 512 feet.

$$192t - 16t^2 = 512.$$

$$16t^2 - 192t + 512 = 0$$

$$t = 4, \ t = 8. \rightarrow (1)$$

On the way up ($t = 4$),

velocity, $v = 192 - 32(4) = 64$ ft/sec

speed $= |v| = 64$ ft/sec $\rightarrow (1)$

On the way ~~up~~ down ($t = 8$)

velocity, $v = 192 - 32(8) = -64$ ft/sec $\rightarrow (1)$

speed $= |v| = 64$ feet/sec.

(c) The rock is aloft for $6 + 6 = 12$ sec $\rightarrow (1)$